MAT113 – Mathematics for Biosciences

B.Sc. Forensic Science, Semester I (Batch 2026–2029)

Unit II Assignment: Set Theory

Practice Questions-2

Part A: Set Notation and Types

Q1. Write the following in **roster form**:

$$\{x \in \mathbb{N} \mid x < 8,\ x \text{ is odd}\}$$

Q2. Let $A = \{a, b\}$. Write all the possible subsets of A.

Part B: Union, Intersection, and Difference

Q3. $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7, 8\}$. Find $A \cup B$, $A \cap B$, $A - B$, $B - A$.

Q4. $A = \{a, e, i, o, u\}$, $B = \{a, b, c, d, e\}$. Find $A \cup B$, $A \cap B$, $A - B$, $B - A$.

Q5. $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{3, 4, 5, 6, 7\}$. Find A', B', $(A \cup B)'$, $(A \cap B)'$.

Q6. With $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$. **Verify De Morgan's Law 1:** $(A \cup B)' = A' \cap B'$.

Part C: Venn Diagrams

Q7. $A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$. Draw a Venn diagram and shade $A \cup B$.

Q8. Draw a Venn diagram and shade $(A \cup B)'$.

Q9. Draw a 3-set Venn diagram with sets A, B, C and shade $A \cap B \cap C$.

Part D: Counting of Sets

Q10. If $n(A) = 15$, $n(B) = 20$, $n(A \cap B) = 8$, find $n(A \cup B)$.

Q11. If $n(U) = 50$, $n(A) = 32$, find $n(A')$.

Q12. In a class of 40 students, 25 play cricket, 20 play football, 10 play both.

(a) How many play at least one sport?

(b) How many play only cricket?

(c) How many play only football?

(d) How many play neither?

Part E: Inclusion-Exclusion

Q13. 150 students: 90 like pizza, 70 like burger, 50 like both.

(a) How many like at least one?

(b) How many like only pizza?

(c) How many like neither?

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Q14. In a group of 50 people, 30 read Hindi newspaper, 25 read English newspaper, 10 read both.

(a) How many read at least one newspaper?

(b) How many read exactly one?

(c) How many read neither?

Part F: Cartesian Product

Q15. $A = \{1, 2\}$, $B = \{a, b, c\}$. Find $A \times B$ and $B \times A$. Are they equal?

Q16. If $(2x - 1,\ y + 3) = (5, 7)$, find x and y.

Part G: Relations

Q17. On $A = \{1, 2, 3\}$, $R = \{(1, 1), (2, 2), (3, 3), (1, 2)\}$.

(a) Create Arrow diagram of the relation.

(b) Domain R ?

(c) Range R ?

Q18. On $A = \{1, 2, 3, 4, 5\}$, $R = \{(a, b) \in A \times A : (b - a) iseven\}$. List all elements of R. Is R a function? Justify.

Part H: Relation is a Function?

Q19. $A = \{1, 2, 3\}$, $B = \{a, b, c\}$, $R = \{(1, a), (1, b), (2, c), (3, a)\}$. Is R a function? Explain.

Part I: Mixed / Challenge

Q20. In a class of 50 students:

- 30 passed Maths
- 25 passed Physics
- 15 passed both

(a) Draw a labelled Venn diagram.

(b) Find the number who passed only Maths.

(c) Find the number who passed only Physics.

(d) Find the number who passed neither.

— End of Assignment —